

Jiale Zhang

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RESEARCH INTEREST

My research focuses on modeling *vibration sensing and its integration with multimodal systems*. I work at the intersection of physics-based modeling and data-driven machine learning, designing sensing frameworks where excitation, measurement, and analysis iteratively refine one another. A central part of my approach is iterative validation between lab prototyping and real-world deployment. This cycle enables robust, scalable systems that transform everyday structures into reliable sensing platforms.

EDUCATION

Present	Ph.D. in Electrical and Computer Engineering , University of Michigan, Ann Arbor Major GPA: 3.9/4.0 Advisor: Prof. Pei Zhang & Prof. Junyi Zhu
Dec 2022	M.S. in Electrical and Computer Engineering , University of Michigan, Ann Arbor Major GPA: 3.9/4.0 Advisor: Prof. Alanson Sample
July 2020	B.E. in Electronic Information Engineering , ShanghaiTech University, Shanghai Major GPA: 3.9/4.0

PUBLICATIONS

CONFERENCE PAPER:

- 2026 [C.3] Wang, Ruiqing, [Jiale Zhang](#), Sungmin Lee, Jeremy Miles, Gary Rohrer, and Yiwen Dong. **"Behav2Need: Ethology-Driven Animal Need Understanding via Multimodal Sensing on Smart Farms."** In *Proceedings of the 13th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, pp. 13-23. 2026.
- 2025 [C.2] Chang, Yen Cheng, Jesse Codling, Yiwen Dong, [Jiale Zhang](#), Hae Young Noh, and Pei Zhang. **"ViLA: Leveraging General-Purpose Audio for Training Vibration-Based Stadium Crowd Monitoring Models."** In *Proceedings of the 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, pp. 96-106. 2025.
-  **Best Paper**
- 2025 [C.1] Codling, Jesse R., Jeffrey D. Shulkin, Yen-Cheng Chang, [Jiale Zhang](#), Hugo Latapie, Hae Young Noh, Pei Zhang, and Yiwen Dong. **"Flohr: Ubiquitous Heart Rate Measurement Using Indirect Floor Vibration Sensing."** In *Proceedings of the 11th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, pp. 44-54. 2024.
-  **Best Paper Runner-up**

JOURNAL PAPER:

- 2026 [J.5] Zhang, Jiale, Shufeng Yin, Lingxiao Yang, Jisu Yim, Kaiyi Guo, Yuxuan Miao, Daniel Vena, Pei Zhang, Yiyue Luo, Junyi Zhu. **"SleepVibe: Sleep Stage Monitoring Through Active Bed-Frame Vibration."** *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2026 (In press)
- 2026 [J.4] Codling, Jesse R., Julia Gersey, Jeffrey D. Shulkin, Yen-Cheng Chang, Jiale Zhang, Hugo Latapie, Hae Young Noh, Pei Zhang, and Yiwen Dong. **"FloorPulse: Heart rate measurement at a distance using indirect floor vibrations."** *Data-Centric Engineering* 7 (2026): e36.
- 2022 [J.3] Zhang, Jiale, Chenzhe Li, Weichao Jiang, Zhicheng Wang, Lejia Zhang, and Xiong Wang. **"Deep-learning-enabled microwave-induced thermoacoustic tomography based on sparse data for breast cancer detection."** *IEEE Transactions on Antennas and Propagation* 70, no. 8 (2022): 6336-6348.
- 2020 [J.2] Zhang, Dajun, Zhansong Lin, Ji Liu, Jiale Zhang, Zhengping Zhang, Zhang-Cheng Hao, and Xiong Wang. **"Broadband high-efficiency multiple vortex beams generated by an interleaved geometric-phase multifunctional metasurface."** *Optical Materials Express* 10, no. 7 (2020): 1531-1544.
- 2020 [J.1] Jiang, Weixiong, Heng Yu, Jiale Zhang, Jiaxuan Wu, Shaobo Luo, and Yajun Ha. **"Optimizing energy efficiency of CNN-based object detection with dynamic voltage and frequency scaling."** *Journal of Semiconductors* 41, no. 2 (2020): 022406.

BOOK CHAPTERS:

- 2026 [B.1] Wu, Yuyan, Jiale Zhang, Moon Lee, Cherrelle Smith, Pei Zhang, and Hae Young Noh. **"Body weight estimation through bed structural vibrations."** In *Dynamics of Civil Structures*, Vol. 2, pp. 147-155. River Publishers, 2026.

EXTENDED ABSTRACTS/POSTER& WORKSHOP PAPER:

- 2025 [P.13] Zhang, Jiale*, Linglong Qian*, Yi Rong*, Junyi Zhu, and Pei Zhang. **"WiP: FaceTap—On-Face Tap Interaction Using Bone-Conducted Vibrations."** *The 32nd Annual International Conference on Mobile Computing and Networking (MobiCom '26)*, 2026. (In press)
- 2025 [P.12] Yin, Shufeng*, Jiale Zhang*, Lingxiao Yang, Yi Rong, Linglong Qian, Pei Zhang, and Junyi Zhu. **"ABreath: Bed-Frame-Mounted Vibration Sensing for Noise-Robust Abnormal Breathing Detection."** *The 4th International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys'26)*, 2026. (In press)
- 2025 [P.11] Fernandez, Tomas Samuel, Yen Cheng Chang, Jesse Codling, Jiale Zhang, Julia Gersey, Nate Sochocki, Sravya Vahini Ganti, Jatin Aggarwal, JeongGil Ko, Hae Young Noh, Lars Junghans, Pei Zhang. **"HotRGB: Temperature Aware RGB-to-Thermal Training Data for Drive-by Urban People Detection."** *The 4th International Workshop on Human-Centered Sensing, Modeling, and Intelligent Systems (HumanSys'26)*, 2026. (In press)
- 2025 [P.10] Zhang, Jiale, Yuyan Wu, Jesse Codling, Julia Gersey, Adeola Bannis, Carlos Ruiz Dominguez, Ke Sun, and Pei Zhang. **"On-Shelf Weight Difference Estimation Through**

Active Vibration Sensing." In *Proceedings of the 23rd ACM Conference on Embedded Networked Sensor Systems*, pp. 640-641. 2025.

- 2025 [P.9] Gersey, Julia, Troy Zhong, Jiale Zhang, Jesse Codling, Jackelyn Hwang, and Pei Zhang. **"Human vs. Machine: Comparing Urban Condition Classification Methods from Vehicular Vision."** In *Proceedings of the 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, pp. 427-430. 2025.
- 2025 [P.8] Gersey, Julia, Jatin Aggarwal, Jiale Zhang, Jesse Codling, and Pei Zhang. **"Sniffing Out the City-Vehicular Multimodal Sensing for Environmental and Infrastructure Analysis."** In *Proceedings of the 23rd ACM Conference on Embedded Networked Sensor Systems*, pp. 632-633. 2025.
-  **Best Poster**
- 2025 [P.7] Chang, Yen Cheng, Jesse Codling, Yiwen Dong, Jiale Zhang, Jiasi Chen, Hae Young Noh, and Pei Zhang. **"Leveraging General-Purpose Audio Datasets for Vibration-based Crowd Monitoring in Stadiums."** In *Proceedings of the 23rd ACM Conference on Embedded Networked Sensor Systems*, pp. 590-591. 2025.
- 2024 [P.6] Fernandez, Tomas, Yen Cheng Chang, Jesse Codling, Yiwen Dong, Jiale Zhang, Carlee Joe-Wong, Hae Young Noh, and Pei Zhang. **"Poster: Drive-by City Wide Trash Sensing for Neighborhood Sanitation Need."** In *Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services*, pp. 704-705. 2024.
- 2024 [P.5] Chang, Yen Cheng, Jesse Codling, Yiwen Dong, Jiale Zhang, Jeffrey Shulkin, Hugo Latapie, Carlee Joe-Wong, Hae Young Noh, and Pei Zhang. **"Listen and Then Sense: Vibration-based Sports Crowd Monitoring by Pre-training with Public Audio Datasets."** In *2024 23rd ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN)*, pp. 285-286. IEEE, 2024.
- 2024 [P.4] Codling, Jesse R., Jeffrey D. Shulkin, Yiwen Dong, Jiale Zhang, Hugo Latapie, Hae Young Noh, and Pei Zhang. **"Demo abstract: FreePulse heart rate monitoring system using ambient structural vibrations."** In *Proceedings of the 22nd International Conference on Information Processing in Sensor Networks*, pp. 364-365. 2023.
- 2023 [P.3] Zhang, Jiale, Shweta Pati, Jesse Codling, Adeola Bannis, Carlos Ruiz, Hae Young Noh, and Pei Zhang. **"Vibration-Based Object Classification with Structural Response of Ambient Music."** In *Proceedings of the 22nd International Conference on Information Processing in Sensor Networks*, pp. 314-315. 2023.
- 2021 [P.2] Zhang, Jiale. **"Directly Controlling the Perceived Difficulty of a Shooting Game by the Addition of Fake Enemy Bullets."** In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*, pp. 1-5. 2021.
- 2019 [P.1] Zhang, Dajun, Jiale Zhang, Ji Liu, and Xiong Wang. **"Topological transformation OAM based on metasurface generated Bessel beam."** In *2019 International Symposium on Antennas and Propagation (ISAP)*, pp. 1-2. IEEE, 2019.

Under Review/ Preprints:

2026 [U.6]	Julia Gersey, Jeongmin Chae, <u>Jiale Zhang</u> , Urbashi Mitra, Pei Zhang. “ RECOVER: Matrix-Based Reconstruction of Missing Vibration Sensor Signals for Robust Activity Recognition ”, Under Review
2026 [U.5]	Yuyan, Wu, <u>Jiale Zhang</u> , Moon Lee, Cherrelle Smith, Xinyi Li, Ankur Senapati, Pei Zhang, Hae Young Noh. “ MelodyBedScale: Human Body Weight Estimation Through Music-Induced Bed Vibrations ”, Under Review
2026 [U.4]	<u>Jiale Zhang</u> , Shufeng Yin, Lingxiao Yang, Amanda Xia, Kaiyi Guo, Yuyan Wu, Daniel Vena, Pei Zhang, Yiyue Luo, Junyi Zhu. “ ActiFlow: Active Bed-Frame Vibration Sensing for Sleep Respiratory Airflow Monitoring. ” Under Review
2026 [U.3]	Kaiyi Guo, Shufeng Yin, <u>Jiale Zhang</u> , Yizhe Shen, Junyi Zhu. “ RespiraFrame: Respiratory Monitoring via Electrical Impedance Tomography and Bone-Conduction Acoustics on Glasses Frame. ” Under Review
2025 [U.2]	Wang, David, Derek Goh, and <u>Jiale Zhang</u> . "Material Identification Via RFID For Smart Shopping." <i>arXiv preprint arXiv:2504.17898</i> (2025).
2025 [U.1]	<u>Zhang, Jiale</u> , Yuyan Wu, Jesse R. Codling, Yen Cheng Chang, Julia Gersey, Pei Zhang, Hae Young Noh, and Yiwen Dong. "WeVibe: Weight Change Estimation Through Audio-Induced Shelf Vibrations In Autonomous Stores." <i>arXiv preprint arXiv:2502.12093</i> (2025).

FUNDING

2026	Active Vibration Sensing for Food Quality Monitoring in Smart Refrigerator 100,000\$. Springhouse Technologies Inc.
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WORK EXPERIENCE

May-Aug 2025	PhD Research Intern , Dolby Laboratories, San Francisco, CA Project: Intrinsic and Extrinsic Feedback Design for Vocal Training System on Glasses Advisor: Mark Thomas
May-Aug 2024	AI Engineer Intern , Aifi Inc., Ann Arbor, MI Project: Item-Customer Association by Camera-RFID Fusion in Autonomous Stores Advisor: Andrew Merrow

TEACHING EXPERIENCE

2025 [T.3]	EECS507: Introduction to Embedded System Research. GSI. EECS, Umich.
2024 [T.2]	EECS215: Introduction to Circuit Basics. GSI. EECS, Umich.
2022 [T.1]	EECS507: Introduction to Embedded System Research. GSI. EECS, Umich.

HONORS AND AWARDS

2025	Center for Pandemic Insights (CPI) Trainee Transdisciplinary Research Awards
2023	Qualcomm Innovation Fellowship

2021 **Rackham International Student Fellowship**

2018 **First Prize in Second Shanghai Maker Contest**

SKILLS

Programming Language: Python, C/C++, C#, Verilog/System Verilog/VHDL, MATLAB

Development Kit: PyTorch, Quartus, Vitis, ESP32, STM32, Labview, Fusion360